

DEPARTMENT OF HOMELAND SECURITY
UNITED STATES COAST GUARD
SHORE INFRASTRUCTURE MAINTENANCE COMMAND

SPECIFICATIONS FOR
REPLACE MEDIUM VOLTAGE CUTOFF SWITCHES
PSN 16677450

AT

U. S. COAST GUARD
AIR STATION CAPE COD
OTIS ANGB, MASSACHUSETTS
(Buzzards Bay)

J. Tessier
PROJECT ENGINEER

David Dill, PE
BRANCH CHIEF



MICHAEL P. CAROSOTTO, PE

TECHNICAL DIRECTOR
UNITED STATES COAST GUARD
CIVIL ENGINEERING UNIT PROVIDENCE
475 KILVERT STREET
WARWICK, RHODE ISLAND 02886

PROJECT TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

01 11 00	SUMMARY OF WORK
01 14 00	WORK RESTRICTIONS
01 32 01	PROJECT SCHEDULE
01 33 00	SUBMITTAL PROCEDURES
01 50 00	TEMPORARY FACILITIES AND CONTROLS
01 52 50	SAFETY REQUIREMENTS
01 57 19	TEMPORARY ENVIRONMENTAL CONTROLS
01 77 00	CLOSEOUT PROCEDURES
01 78 23	OPERATION AND MAINTENANCE DATA
01 80 00	FACILITIES PREVENTATIVE MAINTENANCE PROGRAM

-- End of Project Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

1.1 WORK COVERED BY CONTRACT DOCUMENTS

1.1.1 Project Description

- a. The Coast Guard requires design-build services to provide replacement of (2) 5KV distribution switches and all, incidental wiring and incidental related work. The New switches will be replaced with a new state-of-the-art system that meets current state and federal codes, design standards and industry best practices.
- b. WORK SCOPE: Contractor shall furnish all design, labor, materials, equipment, supervision, transportation and disposal necessary to provide a complete operational 5KV distribution system that meets or exceeds the National Electrical Code. The work shall include, but is not limited to, the following:
 1. Remove and dispose of (2) existing 5KV SF6 GAS PUFFER medium voltage cutoff switches and their cabinets.
 2. Provide (2) new NEMA 4X enclosures 5KV distribution switches to replace existing.
 3. Cutoff switch 86936 services the transformer that provides power to buildings 3161, 3162, 3163, and transformer that provides power to buildings Hangar 3172.
 4. Cutoff switch 86937 services the transformer that provides power to buildings 3159 and 3154.
 5. The switches and, all materials, shall be disposed of by incineration and in accordance with federal and state regulations. See specification for additional requirements. See specification 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS for additional information and requirements.
 6. The contractor shall Remove and dispose of (2) existing concrete pad and provide (2) new concrete pads sized as required to support the replacement of new 5KV switchgear.
 7. If Earth disturbance in previously undisturbed areas or at depths that are previously undisturbed requires consultations with the CEUP's NEPA Environmental Reviewer. Contractor shall report to COR before any Earth disturbance.
 8. Power outages shall be limited to a maximum (4) hours per 24-hour period. If more time is needed, a generator shall be provided for powering of all

- facilities affected during site power interruptions. for the duration of the power outage. The contractor shall supply all connections and fuel for temporary generator. The contractor shall notify the unit 14 days prior to Power outages to the buildings. The contractor shall ensure connections are secured and safeguarded for the safety of unauthorized personnel.
9. The contractor shall provide a detailed plan to the Coast Guard that includes the temporary generator size, expected fuel consumption, expected fuel capacity, load calculations, and an after hours service representative for emergency maintenance of generator in case of unexpected failure. This service representative shall respond to phone call within one hour and shall be on site within 3 hours of emergency call.
 10. Upon completion of each work day, the contractor shall ensure there are no exposed live parts and secure the entire construction area via a locked, 6' high chain link fence, with warning signs, for the safety of unauthorized personnel.
- c. LOCATION: USCG CG Base Cape Cod Bldg 5201, Joint Base Cape Cod, 3159 Burge Blvd, Buzzards Bay, MA 02542.
- d. REQUIRED SERVICES: The renewed new 5KV distribution switches system shall have a minimum design life expectancy of 20 years (using commercial and industrial grade equipment) and provide maintenance personnel with a service friendly system.
1. The contractor shall submit and utilize a quality control plan to assure quality requirements are being met. The contractor shall review all submitted design documentation for completeness and accuracy.
 2. The contractor shall be responsible for start-up and test of all systems in accordance with the equipment manufacturer and the NFPA testing guidelines. The testing shall include start-up, performance verification, testing, and adjusting.
 3. The contractor shall utilize a licensed Electrical engineer with specialized experience with utility voltage to design and prepare installation/layout drawings for all work. The engineer shall conduct submittal reviews (if required), to ensure compliance with design drawings and shall certify all tests, including final systems test. The licensed engineer shall provide the Contracting Officer with a signed

certificate that the systems were installed and tested in accordance with approved plans.

4. All electrical work shall be accomplished by qualified Massachusetts licensed electrician. The electrician shall be experience with medium voltage electrical systems. The contractor shall submit a copy of the master electrician's license/certification for review to the Contracting Officer before beginning any electrical work.
 5. All work shall be in strict accordance with the NFPA 70 (National Electrical Code) and IEEE (ANSI C2), with special focus shall be on grounding and bonding. All work shall be reviewed and approved by a registered professional electrical engineer experienced with medium voltage and standby generator systems.
 6. Work and materials shall comply with applicable laws, ordinances, rules and regulations, including national, state, and local electrical codes. As a minimum, materials and installation shall comply with NFPA.
- e. All buildings will remain in operation throughout the project. The contractor shall ensure all areas are protected during construction and kept clean at the end of each day.
- f. REFERENCE DOCUMENTS: AutoCAD drawings will be provided as reference to the original equipment installation. The contractor shall verify all existing conditions. These drawings are diagrammatic only and show the general area occupied which may be different than the actual field conditions. The existing equipment and associated wiring within these drawing are shown diagrammatically and may not include all associated equipment to be replaced. The contractor shall field verify all existing conditions.

Title

- | | |
|---|------------------------|
| 1 | COVER SHEET |
| 2 | SITE PLAN |
| 3 | LARGE SCALE SITE VIEWS |
| 4 | DETAILS |
| 5 | DIAGRAM |

- g. PREPARE DRAWINGS: The contractor shall provide design drawings for all work, including but not limited to existing conditions, demolition work and new work, to the

Contracting Officer or duly appointed representative for review and approval. The design shall incorporate, as a minimum, all the requirements specified herein. Provide plan(s) showing the chosen manufacturer's data, functional layouts of equipment, equipment location, sizes, underground conduit runs and connections. Provide elevations, sections and details where necessary for clarity. Provide complete schedule for all equipment. Provide legend, symbols and abbreviations applicable to the work. Provide all design calculations and provide a concise summary of all the calculations. The Contractor shall furnish equipment that is new and a standard product of the chosen manufacturer and currently commercially available. The models quoted shall include the latest engineering changes of the manufacturer.

- h. CADD Drawings: Submittal and final construction drawings shall be on 22" X 34" paper with a drawing number as assigned. Drawings shall be layers, lettering, line weight, symbols, and schedules shall be in conformance with the USACE A/E/C CAD Standards. All X-refs shall be bound to the original drawings prior to the final submission.
- i. The Contractor is encouraged to contact the CEU Providence CAD Manager for assistance in preparation of design drawings. USCG CAD Standard/Drawing Format and Handling Job Guide will be provided. Each drawing shall have its own file and file name, with format as directed by CEU Providence CAD Manager and Design Project Manager. Progress and final drawings submitted will be electronically reviewed for compliance with the USCG CAD Standard
- j. Drawings shall be completed using software capable of producing an Autodesk AutoCAD Release 2020 or earlier compatible .dwg file format deliverable. They shall also conform to the latest USCG CAD, while using the standard USCG drawing border template (USCG provided).
- k. Contractor is responsible for providing a complete and operational 5KV system in strict accordance with ALL codes, standards, manufacturer's instructions and this written specification. If there is a conflict in the code requirements, the more strict interpretation shall take precedence. Upon completion of the system installation, the system shall be operated, tested and approved by the manufacturer's representative. The manufacturer's representative shall sign off, in writing, stating the system meets their installation and operating instructions and that the system meets the manufacturer's warranty. The contractor shall incur all costs for these

services. Contractor is responsible for ALL incidental related work.

- l. The Coast Guard review of the design does not constitute approval or acceptance of any variation from the RFP, unless variations have been specifically pointed out in writing by the contractor and specifically approved in writing by the contracting officer.
- m. All final design drawings and calculations shall be signed, dated, and bear the seal of the licensed engineer.

1.2 DEFINITIONS

The following terms, when used in the Contract Documents, have the meaning described.

- a. Contracting Officer
The U.S. Coast Guard's Contracting Officer.
- b. Contracting Officer's Representative (COR)
The Coast Guard's assigned technical representative.
- c. Contractor
The word "Contractor" is also deemed to include a Design-Build Firm contracting with the U.S. Coast Guard for performance of work, including all engineering and construction services as well as the furnishing and installation of materials.
- d. Construction Superintendent
Design-Build firms on site project supervisor responsible for overseeing all tradesperson and to ensure that the construction is in compliance with the contract documents as well as assuring that all on-base regulations are followed. This person will also be the on site point of contact for the Coast Guard's Project Manager, Inspector, and personnel.
- e. Design-Build (D-B)
Design-Build means combining into a single contract the project's design and construction phases, including construction engineering and inspection services.
- g. Designer of Record (DOR)
The Professional Engineer that develops the concept and design for the project, performs the analysis and engineering calculations, and is responsible for the preparation of the following: Construction Documents; answering all design

and/or construction questions pertaining to the design; approval authority for technical submittals in conjunction with the Government Project Manager; final inspection and generation of "punch list" items that do not meet the design documents and/or approved submittals.

h. Project Manager/COR

The U.S. Coast Guard representative that is responsible for the administration of the Design-Build Contract.

i. Inspector

The U.S. Coast Guard representative that is responsible for the oversight of construction activities to ensure compliance of the project documents. The inspector may request that contractors provide manufacturer information of products being used during construction to ensure that the construction documents are being complied with. The Coast Guard Inspector may also inspect any materials that are on site as well as the installation to ensure compliance with the construction and contract documents.

1.3 DESIGN REQUIREMENTS

a. The Contractor shall make every attempt to design around existing utilities, minimizing impacts. This shall require the Contractor to contract a reputable and qualified utility locating service company to identify and clearly mark any existing underground utilities prior to the start of design and construction phases.

d. Upon Notice to Proceed (NTP), the Contractor must meet the following schedule. This shall be considered for bidding purposes:

- i. 65% Design Submission for CG Review: 25 working days from NTP. CEU Providence will have 15 working days to review and comment on this submission.
- ii. 95% Design Submission for CG Review: 65 working days from NTP. CEU Providence will have 15 working days to review and comment on this submission.
- iii. 100% Stamped Drawings and Specifications: 90 working days from NTP.
- iv. Mobilize for the start of construction 14 working days upon CG acceptance of 100% drawing and specification.

- v. If a second edit is required to the 100% design submittals, an additional 5 working days will be allowed for final edits. Any errors/omissions in the final design submission shall be corrected at the construction phase of the project at the contractor's expense. Any corrections shall be indicated on the as-built drawings.
- e. Designer of Record shall be responsible for reviewing, rejecting, and/or approving all technical submittals. However, the U.S. Coast Guard shall also have the opportunity to review and comment on all submittals; both technical and contractual.
- f. Designer of Record shall be available to answer any and all design and construction questions that the Government may have pertaining to the project design documents.
- g. Any proposed changes to the design during construction must first be requested in writing to the Designer of Record as well as the Contracting Officer and approved by the Designer of Record as well as the Contracting Officer prior to executing the change.
- h. Coordinate a final inspection involving all pertinent parties such as representatives of the US Coast Guard. The Designer of Record shall be present for the final inspection and provide written acceptance or a "punch list" of work to be completed and or re-worked prior to the final acceptance. If a punch list is generated by either the DOR and/or the Government during the final inspection, follow up inspections by all parties shall be performed until the work is deemed satisfactory by both the Government and the Designer of Record.

1.4 ADDITIONAL REQUIREMENTS

- a. The Contracting Officer's Representative COR may request that Contractors provide information of products being used during construction to ensure that the construction documents are being complied with. The COR may also inspect any materials that are on site as well as the installation to ensure compliance with the construction and contract documents. This information shall include, but is not limited to, construction methods and materials, the manufacturer's name, trade name, place of manufacture, catalog model or number, nameplate data, size, layout dimensions, capacity specification and technical paragraph reference, UL numbers, installation procedures/instructions, and manufacturer's certifications of all devices, components, and equipment that are being installed including boxes and raceways. Installation of the item shall not proceed until manufacturer's instructions are received. Failure to submit manufacturer's instructions shall be cause for rejection of the equipment or material. The following

additional Submittals requirements shall be include above that of Section 01330, "Submittal Procedures".

b. The Contractor shall provide an on-site, full-time construction superintendent that is English speaking responsible for coordinating all trades and acting as the direct point of contact for the U.S. Coast Guard's Contracting Officer and their on-site technical representative.

c. The construction superintendent shall ensure that all employees and sub-contractors strictly obey the requirements of the U.S Coast Guard, local jurisdictions, and ensure that all security regulations are strictly followed. At no time shall the Contractor, their employees or sub-contractors enter any building without prior approval of the on site representative. Any security violations or on-base violations may lead to that person or persons being restricted from future access to the base.

d. The construction superintendent shall be available at all times to answer any on-site questions from the Contracting Officer and/or the Contracting Officer's Representative (COR). This shall include but is not limited to construction methods and materials. At any time during construction, the on-site representative shall provide materials information for any product being used under this contract and allow access to all areas of construction and/or equipment being used under this contract.

1.5 EXISTING WORK

In addition to "FAR 52.236-9, Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements".

a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work, which remain.

b. Repair or replace portions of existing work, which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work shall be in a condition equal to or better than that which existed before new work started.

1.6 PRECONSTRUCTION CONFERENCE

After award of the contract but prior to commencement of any work at of the sites, meet with the Contracting Officer to discuss and develop a mutual understanding relative to the administration of the value engineering, preparation of the schedule, and other submittals, scheduling programming, and prosecution of the work. Major subcontractors who will engage in the work shall also attend.

1.7 INSPECTION

The Contractor shall keep the Contracting Officer or duly appointed representative fully informed of contract operations and plans so that a representative may arrange to be present at various times that work is being performed.

1.8 Contractor's Daily Log

The Contractor shall complete the USCG Daily Log form at the end of each working day and deliver it to the Government Inspector. The Government Inspector will attest to its accuracy by initialing the form and forwarding that form to CEU Providence. The form will be provided to the Contractor by the Contracting Officer's Representative.

1.9 Provide Operations and Maintenance(O&M) Documentation

The contractor shall provide complete, bound and organized O&M instructions explaining the requirements of the system, upon contract completion.

1.10 BUY AMERICAN ACT COMPLIANCE

All equipment purchased under this contract must comply with the "Buy American Act" (BAA).

1.2 Activity Regulations

Ensure that Contractor personnel employed on Base Cape Cod become familiar with and obey Base Cape Cod regulations including safety, fire, traffic, and security regulations. These include, but are not limited to, the following:

1. Adhere to Base Cape Cod policies for Base access. Provide access applications on forms provided by the Government 3 to 5 days in advance of arrival.
2. Keep within the limits of the work and avenues of ingress and egress.
3. Wear steel toed shoes, long pants, shirts and hard hats in designated areas.
4. Do not enter any restricted areas unless required to do so and until cleared for such entry.
5. The Contractor's equipment shall be conspicuously marked for identification.
6. The use of metal track vehicles is prohibited on paved surfaces.
7. Utilize rubber tire or rubber track vehicles.

Coordinate access to each facility and the project work areas with the COR. Limited on-site parking is available as delineated on the Staging Plan. Car pool and otherwise plan accordingly. Schedule all deliveries with the COR.

Each Contractor's employee must carry a personal photo ID anytime they are on the site.

Submit a typewritten list on company letterhead of all Contractor's related vehicles (work and personal) that will be left unattended within the facility to the COR.

Do not enter private rooms/areas (including offices and Berthing Rooms) without a Government escort.

The installation of any unauthorized electric, electronic, or any other type of monitoring device is strictly forbidden. Any violation of this directive will result in immediate removal of the person(s) responsible, and their employer, from the project for an indefinite period of time, and legal action against the parties involved.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 14 00

WORK RESTRICTIONS

PART 1 GENERAL

1.1 SPECIAL SCHEDULING REQUIREMENTS

- a. Have materials, equipment, and personnel required to perform the work at the site prior to the commencement of the work.
- b. The Contractor shall conduct his operations so as to cause the least possible interference with normal operations of the activity.
- c. Permission to interrupt any roads and/or utility service shall be requested in writing a minimum of 15 calendar days prior to the desired date of interruption.
- d. The work under this contract requires special attention to the scheduling and conduct of the work in connection with existing operations. Identify on the construction schedule each factor which constitutes a potential interruption to operations.

1.2 CONTRACTOR ACCESS AND USE OF PREMISES

1.2.1 Regulations

Ensure that Contractor personnel employed at the sites become familiar with and obey the regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Do not enter any restricted areas unless required to do so and until cleared for such entry. The Contractor's equipment shall be conspicuously marked for identification.

1.2.2 Working Hours

Regular working hours shall consist of an 8 1/2 hour period, between 7:00 a.m. and 3:30 p.m., Monday through Friday, excluding Government holidays. Work outside those hours must be made in writing to the Contracting Officer a minimum of 7 days in advance.

1.2.3 Occupied and Existing Buildings

The Contractor will be working in an existing building which is occupied. Do not enter the building without prior approval of the Contracting Officer. Contractor shall not enter any space without a Coast Guard Representative escort. Only one Coast Guard Representative escorts will be supplied at any one time and no escort will be provided for 1 hour at lunch time.

The existing buildings and their contents shall be kept secure at all times.

1.3 SECURITY REQUIREMENTS

1.3.1 Parking

Do not park on or block the marked fire lanes or crane rail traveling zones
(marked in yellow) at any time.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 32 00

PROJECT SCHEDULE
02/15

PART 1 GENERAL

1.1 SUBMITTALS

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

1.1.1 SD-01 Preconstruction Submittals

Construction Schedule

Schedule of Values

Approval of the above submittals is required prior to the start of construction.

1.2 CONSTRUCTION SCHEDULE AND SCHEDULE OF VALUES

At the Pre-Construction Conference, submit a Construction Schedule and Schedule of Values in accordance with FAR Clause 52.236.15. The Schedule of Values shall be based on the actual breakdown of the bid price. The cost of insurance shall not be listed as a separate item but included as part of each item of work. The actual cost of bonds may be paid as the first progress payment when a receipt from the bonding company is presented to the KO. In addition, keep the Government Inspector informed daily of the expected delivery dates for major pieces of equipment and materials.

1.2.1 Contents

The Construction Schedule and the Schedule of Values shall be incorporated into separate forms which clearly indicate the start and completion dates and unit value of all major work components. Include the following project specific items at a minimum:

- a. Bonds
- b. Pre-Construction Submittals
- c. Mobilization
- d. Work Activities
- e. Factors which Constitute Potential Interruptions to Unit Operations
- f. Potential Interruptions to Unit Operations
- g. [Quality Control]
- h. In-Progress Submittals

- i. Final Government Inspection
- j. Close-Out Submittals; i.e. Testing Balancing Reports, Warranties, Operation and Maintenance Manuals, Posted Instructions
- k. As-Built Drawings
- l. Demobilization

1.2.1.1 Definitions

- a. Pre-Construction Submittals: Pre-Construction Submittals are those activities which encompass the obtaining, submission, review and approval of submittals necessary prior to the start of the related site work. KTR shall annotate all such activities into the progress schedule with their forecasted time periods. These activity periods shall not be exclusive of the contract performance period. No site work shall commence until the respective submittals have been approved.
- b. Mobilization: Mobilization is defined as having equipment AND having a physical presence of at least one person from the KTR's team on the jobsite.
- c. Work Activities: The value of all major work components within the project shall be identified on the Schedule of Values on a unit quantity and unit cost basis, e.g. number of squares and cost per square of roofing, number of lineal feet and cost per lineal foot of conduit, number of panels and cost per.

1.2.2 Format

The Construction Schedule shall be provided in a horizontal Gantt-chart format and prepared using MS PROJECT or approved equal. Include any constraints and work restrictions (e.g., exclusionary periods, holidays, government scheduled events, etc.), phasing, work performed by the government, staging of construction for major portions of the work, and milestones. Subdivide schedule into sufficient number of work activities to accurately display the work schedule, sequence of work, activity duration, and interdependence of each activity. Clearly show the activities of the critical path. The critical path is defined as the longest path.

The Schedule of Values shall be provided on the attached KTR's Schedule of Values Form.

Provide one (1) electronic copy of both the Construction Schedule and Schedule of Values in PDF format as well as one (1) electronic copy of the Construction Schedule in MS PROJECT format prior to the start of construction. Provide updated versions in accordance with paragraph Updated Schedules.

1.2.3 KTR Schedule Responsibilities

The Construction Schedule and Schedule of Values as approved by the Government is not a substitute for quantities conveyed by the specification and drawings and those required for a complete job. Omissions and errors on

the Construction Schedule and Schedule of Values are the responsibility of the KTR. Payments will not be made until the schedule of prices has been submitted to and approved by the KO.

1.2.4 Updated Schedules

Update the Construction Schedule, Schedule of Values, and equipment delivery schedule at monthly intervals or when schedule has been revised. Reflect any changes occurring once the last update. Submit copies of the purchase orders and confirmation of the delivery dates as directed.

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The authorized representative must have a minimum of [2] years of experience scheduling construction projects similar in size and nature to this project with scheduling software that meets the requirements of this specification. Representative must have a comprehensive knowledge of CPM scheduling principles and application. Qualifications shall be submitted upon request of the COR.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.2 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the KO directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the KO may withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made.

3.2 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.2.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an appropriate level of detail will result in its disapproval. The KO will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail:

3.2.2 Procurement Activities

Include activities associated with the critical submittals and their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement

activities are those with an anticipated procurement sequence of over [60] calendar days.

3.2.3 Government Activities

Show Government and other agency activities that could impact progress. These activities include, but are not limited to: approvals, environmental permit approvals by State and/or Federal regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

3.2.4 Phase of Work

Assign Phase of Work to all activities. Examples of phase of work are [procurement phase and] construction phase.

3.2.5 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the KO. Propose logic corrections to eliminate out of sequence progress or justify not changing the sequencing for approval prior to submitting an updated project schedule.

3.2.6 Added and Deleted Activities

Do not delete activities from the project schedule or add new activities to the schedule without approval from the KO. Activity number and description changes are considered new activities and cannot be changed without KO approval.

3.2.7 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value.

3.2.8 Remaining Duration

Update the remaining duration for each activity based on the number of estimated work days it will take to complete the activity.

3.3 REQUESTS FOR TIME EXTENSIONS

Provide a justification of delay to the KO in accordance with the contract provisions and clauses for approval within 10 days of a delay occurring. Also prepare a time impact analysis for each Government request for proposal (RFP) to justify time extensions.

3.3.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date or interim completion date(s).

3.3.2 Time Extension

No time extension will be granted without KO approval of the Justification of Delay and time impact analysis. Approval will not be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the KTR's own actions, which result in a calculated schedule delay will not be a cause for an extension to the performance period, completion date, or any interim milestone date.

3.4 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the KO may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra work days, etc.

3.4.1 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the KO pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.5 MODIFICATIONS

When a modification is issued by the Government, record the modification as the last activity of the Construction Schedule and Schedule of Values and include the value of the modification. Adjust the Schedule of Values to reflect the inclusion of the modification. Revise the Construction Schedule portion of the form to annotate the progress change. Enter all modifications in this manner in sequential order.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Submittal

The KO may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

1.1.2 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and

1.1.2 Types of Submittals

All submittals are classified as indicated in paragraph "Submittal Descriptions (SD)". Submittals also are grouped as follows:

- a. Shop drawings: As used in this section, drawings, schedules, diagrams, and other data prepared specifically for this contract, by contractor or through contractor by way of subcontractor, manufacturer, supplier, distributor, or other lower tier contractor, to illustrate portion of work.
- b. Product data: Preprinted material such as illustrations, standard schedules, performance charts, instructions, brochures, diagrams, manufacturer's descriptive literature, catalog data, and other data to illustrate portion of work, but not prepared exclusively for this contract.
- c. Samples: Physical examples of products, materials, equipment, assemblies, or workmanship that are physically identical to portion of work, illustrating portion of work or establishing standards for evaluating appearance of finished work or both.
- d. Administrative submittals: Data presented for reviews and approval to ensure that administrative requirements of project are adequately met but not to ensure directly that work is in accordance with design concept and in compliance with contract documents.

1.1.3 Submittal Descriptions (SD)

SD-01 Preconstruction Submittals

Construction Progress Schedule
Submittal register
Schedule of values
Health and safety plan
Work plan
Quality control plan
Environmental protection plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the contractor for integrating the product or system into the project.

Drawings prepared by or for the contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-05 Design Data

Calculations, mix designs, analyses or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. (Testing must have been within three years of date of contract award for the project.)

Report which includes findings of a test required to be performed by the contractor on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report which includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements signed by responsible officials of manufacturer of product, system or material attesting that product, system or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a supplier, installer or subcontractor through Contractor, the purpose of which is to further quality of orderly progression of a portion of the work by documenting procedures, acceptability of methods or personnel qualifications.

Confined space entry permits.

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and Material Safety Data sheets concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative to confirm compliance with manufacturer's standards or instructions.

SD-10 Operation and Maintenance Data

Data intended to be incorporated in operations and maintenance manuals.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

As-built drawings

Special warranties

Posted operating instructions

Training plan

SD-12 Facilities Preventative Maintenance Program (FPMP)

The KTR shall develop and submit an FPMP for all newly constructed or installed infrastructure that accurately identifies the preventive maintenance required to professionally maintain each system or piece of equipment provided in this contract.

1.1.4 Approving Authority

Person authorized to approve submittal.

1.1.5 Work

As used in this section, on- and off-site construction required by contract documents, including labor necessary to produce construction and materials, products, equipment, and systems incorporated or to be incorporated in such construction.

1.2 SUBMITTALS

Submit the following in accordance with the requirements of this section.

SD-01 Preconstruction Submittals

Submittal register

1.3 USE OF SUBMITTAL REGISTER

Prepare and maintain submittal register, as the work progresses. Do not change data in columns (c), (d), (e), and (f) as delivered by government.

1.3.1 Submittal Register

Submit submittal register. Do not change data in columns (c), (d), (e), and (f) as delivered by the government. Verify that all submittals required for project are listed and add missing submittals. Complete the following on the register:

Column (a) Activity Number: Activity number from the project schedule.

Column (b) Transmittal Number: Contractor assigned list of consecutive numbers.

Column (g) Contractor Submit Date: Scheduled date for approving authority to receive submittals.

Column (h) Contractor Approval Date: Date contractor needs approval of submittal.

Column (i) Contractor Material: Date that contractor needs material delivered to contractor control.

1.4 PROCEDURES FOR SUBMITTALS

1.4.1 Reviewing, Certifying, Approving Authority

Government organization shall be responsible for reviewing and certifying that submittals are in compliance with contract requirements. Approving authority on submittals is the Project Manager unless otherwise specified for specific submittal.

1.4.2 Constraints

- a. Submittals listed or specified in this contract shall conform to provisions of this section, unless explicitly stated otherwise.
- b. Submittals shall be complete for each definable feature of work; components of definable feature interrelated as a system shall be submitted at same time.
- c. When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, submittal will be returned without review.
- d. Approval of a separate material, product, or component does not imply approval of assembly in which item functions.

1.4.3 Scheduling

- a. Coordinate scheduling, sequencing, preparing and processing of submittals with performance of work so that work will not be delayed by submittal processing. Allow for potential requirements to resubmit.
- b. Except as specified otherwise, allow review period, beginning with receipt by approving authority that includes at least 10 working days. Period of review for each resubmittal is the same as for initial submittal.

1.4.4 Variations

Variations from contract requirements require Government approval pursuant to contract Clause entitled "FAR 52.236-21, Specifications and Drawings for Construction" and will be considered where advantageous to government.

1.4.4.1 Considering Variations

Discussion with contracting officer prior to submission, will help ensure functional and quality requirements are met and minimize rejections and resubmittals. When contemplating a variation which results in lower cost, consider submission of the variation as a Value Engineering Change Proposal (VECP).

1.4.4.2 Proposing Variations

When proposing variation, deliver written request to the contracting officer, with documentation of the nature and features of the variation and why the variation is desirable and beneficial to government. If lower cost is a benefit, also include an estimate of the cost saving. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

1.4.4.3 Warranting That Variation Are Compatible

When delivering a variation for approval, contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.4.4.4 Review Schedule Is Modified

In addition to normal submittal review period, a period of 5 working days will be allowed for consideration by the Government of submittals with variations.

1.4.5 Contractor's Responsibilities

- a. Determine and verify field measurements, materials, field construction criteria; review each submittal; and check and coordinate each submittal with requirements of the work and contract documents.
- b. Transmit submittals to Contracting Officer in accordance with schedule on approved Submittal Register, and to prevent delays in the work, delays to government, or delays to separate contractors.
- c. Advise contracting officer of variation, as required by paragraph entitled "Variations."
- d. Correct and resubmit submittal as directed by approving authority. When resubmitting disapproved transmittals or transmittals noted for resubmittal, the contractor shall provide copy of that previously submitted transmittal including all reviewer comments for use by approving authority. Direct specific attention in writing or on resubmitted submittal, to revisions not requested by approving authority on previous submissions.
- e. Furnish additional copies of submittal when requested by contracting officer, to a limit of 20 copies per submittal.
- f. Complete work which must be accomplished as basis of a submittal in time to allow submittal to occur as scheduled.
- g. Ensure no work has begun until submittals for that work have been returned as "approved," except to the extent that a portion of work must be accomplished as basis of submittal.

1.4.6 Government Responsibilities

- a. Note date on which submittal was received from contractor on each Contract Item Approval Request Form.
- b. Review each submittal; and check and coordinate each submittal with requirements of work and contract documents.
- c. Review submittals for conformance with project design concepts and compliance with contract documents.
- d. Ensure that material is clearly legible.

e. Review submittals for approval within scheduling period specified and only for conformance with project design concepts and compliance with contract documents.

f. Identify returned submittals with one of the actions defined in paragraph entitled "Actions Possible" and with markings appropriate for action indicated.

1.4.7 Actions Possible

Submittals will be returned with one of the following notations:

a. Submittals marked "approved" "approved as submitted" authorize contractor to proceed with work covered.

b. Submittals marked "approved" "see below" authorize contractor to proceed with work as noted provided contractor takes no exception to the notations.

c. Submittals marked "disapproved" indicate submittal is incomplete or does not comply with design concept or requirements of the contract documents and shall be resubmitted with appropriate changes. No work shall proceed for this item until resubmittal is approved.

1.5 FORMAT OF SUBMITTALS

1.5.1 Transmittal Form

Transmit each submittal, to office of approving authority. Transmit submittals with transmittal form prescribed by contracting officer and standard for project. The transmittal form shall identify contractor, indicate date of submittal, and include information prescribed by transmittal form and required in paragraph entitled "Identifying Submittals." 4 copies of the submittal are required. Submittals made without proper transmittal form will be returned without being reviewed.

1.5.2 Identifying Submittals

Identify submittals, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

a. Project title and location.

b. Construction contract number.

c. Section number of the specification section by which submittal is required.

d. Submittal description (SD) number of each component of submittal.

e. When a resubmission, alphabetic suffix on submittal description, for example, SD-10A, to indicate resubmission.

f. Name, address, and telephone number of subcontractor, supplier, manufacturer and any other second tier contractor associated with submittal.

g. Product identification and location in project.

1.5.3 Format for Product Data

a. Present (4) product data submittals for each section as a complete, bound volume. Include table of contents, listing page and catalog item numbers for product data.

b. Indicate, by prominent notation, each product which is being submitted; indicate specification section number and paragraph number to which it pertains.

c. Supplement product data with material prepared for project to satisfy submittal requirements for which product data does not exist. Identify this material as developed specifically for project.

1.5.4 Format for Shop Drawings

a. Shop drawings shall not be less than 8 1/2 by 11 inches nor more than 30 by 42 inches.

b. Present 8 1/2 by 11 inches sized shop drawings as part of the bound volume for submittals required by section. Present larger drawings in sets.

c. Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph entitled "Identifying Submittals."

d. Dimension drawings, except diagrams and schematic drawings; prepare drawings demonstrating interface with other trades to scale. Shop drawing dimensions shall be the same unit of measure as indicated on the contract drawings. Identify materials and products for work shown.

1.5.5 Format of Administrative Submittals

a. When submittal includes a document which is to be used in project or become part of project record, other than as a submittal, do not apply contractor's approval stamp to document, but to a separate sheet accompanying document.

b. Operation and Maintenance Manual Data: Submit in accordance with Section 01781, "Operation and Maintenance Data." Include components required in that section and the various technical sections.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of Copies of Product Data

a. Submit four copies of submittals of product data.

1.6.2 Number of Copies of Shop Drawings

Submit shop drawings in compliance with quantity requirements specified for product data.

1.6.3 Number of Copies of Administrative Submittals

a. Unless otherwise specified, submit administrative submittals compliance with quantity requirements specified for product data.

b. Submit administrative submittals required under "SD-19 Operation and Maintenance Manuals" to conform to Section 01781, "Operation and Maintenance Data."

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 50 00

TEMPORARY FACILITIES AND CONTROLS

PART 1 GENERAL

1.1 SUBMITTALS

Submit the following in accordance with Section 01330, "Submittal Procedures."

SD-01 Preconstruction Submittals
Construction site plan

1.2 CONSTRUCTION SITE PLAN

Prior to the start of work, submit a site plan showing the locations of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes used for this contract. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.3 TEMPORARY UTILITIES

1.3.1 Contractor Utilities

The Contractor shall be allowed to use the 120-volt power sources with the exception of communication circuits.

1.4 WEATHER PROTECTION

Take necessary precautions to ensure that roof openings and other openings made by the Contractor are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

1.4.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions shall include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

1.5 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

1.5.1 Interruption of Vehicular Traffic

If during the performance of work, it becomes necessary to modify vehicular traffic patterns at any locations, notify the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a Traffic Control Plan detailing the proposed controls to traffic movement for approval. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic.

1.6 STORAGE AREAS

Contractor shall be responsible for security of his property. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

1.6.1 Storage Size and Location

The site available for storage shall be confined to the indicated operations area as indicated by the Contracting Officer for each site.

1.6.2 Storage in Existing Buildings

The Contractor shall be working in existing buildings; the storage of material will not be allowed in the buildings.

1.7 TEMPORARY BUILDINGS

1.7.1 Trailers or Storage Buildings

Trailers or storage buildings will be permitted, where space is available, subject to the approval of the Contracting Officer. The trailers or buildings shall be in good condition, free from visible damage rust and deterioration, and meet all applicable safety requirements. Trailers shall be roadworthy and comply with all appropriate state and local vehicle requirements. Failure to maintain storage trailers or buildings to these standards shall result in the removal of non-complying units at the Contractor's expense. A sign not smaller than 24 by 24 inches shall be conspicuously placed on the trailer depicting the company name, business phone number, and emergency phone number. Trailers shall be anchored to resist high winds and must meet applicable state of local standards for anchoring mobile trailers.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION 01 52 50

SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241	Safeguarding Construction, Alteration, and Demolition Operations
----------	---

1.2 SUBMITTALS

Submit the following in accordance with Section 01330, "Submittal Procedures."

SD-07 Certificates

Accident Prevention Plan (APP)

1.3 QUALITY ASSURANCE

1.3.1 Meetings

The Safety Officer shall attend the pre-construction conference.

1.3.1.1 New Employee Indoctrination

New employees will be informed of specific site hazards before they begin work. Documentation of this orientation shall be kept on file at the project site.

1.3.2 Certifications

1.3.2.1 Accident Prevention Plan (APP)

Submit the APP at least 15 calendar days prior to start of work at the job site. Make the APP site specific. Notice To Proceed will be given after Government finds the APP acceptable.

1.4 ACCIDENT PREVENTION PLAN (APP)

Prepare the APP. Include the associated AHA and other specific plans, programs and procedures, some of which are listed below.

1.4.1 Contents of the Accident Prevention Plan

- a. Name and safety related qualifications of safety officer (including training and any certifications).
- b. Qualifications of competent and of qualified persons.
- c. Emergency response plan. Include a map denoting the route to the nearest emergency care facility with emergency phone numbers. Contractor may be required to demonstrate emergency response.
- d. Fall Protection and Prevention (FP&P) Plan. The plan shall be site specific and address all fall hazards in the work place. It shall address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 6 feet.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 CONSTRUCTION

Comply with NFPA 241, the accident prevention plan, the activity hazard analysis and other related submittals and activity fire and safety regulations.

3.1.1 Unforeseen Hazardous Material

The design should have identified any hazardous materials such as PCB, lead paint, and friable and nonfriable asbestos. If additional material, not indicated, that may be hazardous to human health upon disturbance during construction operations is encountered, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to "FAR 52.243-4, Changes" and "FAR 52.236-2, Differing Site Conditions."

3.2 PERSONNEL PROTECTION

3.2.1 Hazardous Noise

Provide hazardous noise signs, and hearing protection, wherever equipment and work procedures produce sound-pressure levels greater than 85 dBA steady state or 140 dBA impulse, regardless of the duration of the exposure.

3.2.2 Fall Protection

Enforce use of the fall protection device designated for each specific work activity in the FP&P plan and/or AHA all times when an employee is on a surface 6 feet or more above lower levels. Personal fall arrest systems are required when working from an articulating or extendible boom, scissor

lifts, swing stages, or suspended platform. Fall protection must comply with ANSI A10.14.

3.3 SCAFFOLDING

Employees shall be provided with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Stair towers or ladders built into scaffold systems in accordance with USACE EM 385-1-1 Appendix J are required for work platforms greater than 20 feet in height. Contractor shall ensure that employees that are qualified perform scaffold erection. Do not use scaffold without the capability of supporting at least four times the maximum intended load or without appropriate fall protection as delineated in the accepted fall protection plan. Minimum platform size shall be based on the platform not being greater in height than three times the dimension of the smallest width dimension for rolling scaffold. Some Baker type scaffolding has been found not to meet these requirements. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward. Special care shall be given to ensure scaffold systems are not overloaded. Outrigger brackets used to extend scaffold platforms on self supported scaffold systems for the storage of material is prohibited. The first tie-in shall be at the height equal to 4 times the width of the smallest dimension of the scaffold base.

3.4 ELECTRICAL

3.4.1 Conduct of Electrical Work

Cable intended to be cut must be positively identified and de-energized prior to performing each cut. Positive cable identification must be made prior to submitting any outage request for electrical systems. Arrangements are to be coordinated with the Contracting Officer and Station Utilities for identification. The Contracting Officer will not accept an outage request until the Contractor satisfactorily documents that the circuits have been clearly identified.

3.4.2 Portable Extension Cords

Portable extension cords shall be sized in accordance with manufacturer ratings for the tool to be powered.

3.5 HOUSEKEEPING

3.5.1 Clean-up

All debris in work areas shall be cleaned up daily or more frequently as necessary. Construction debris may be temporarily located in an approved location, however garbage accumulation must be removed each day.

3.5.2 Dust Control

In addition to the dust control measures required elsewhere in the contract documents dry cutting of brick or masonry shall be prohibited. Wet cutting must address control of water run off.

-- End of Section --

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1200	Hazard Communication
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 261	Identification and Listing of Hazardous Waste
49 CFR 173	Shipments and Packagings

1.2 DEFINITIONS

1.2.1 Solid Waste

Garbage, refuse, debris, sludge, or other discharged material (except hazardous waste as defined in paragraph entitled "Hazardous Waste" or hazardous debris as defined in paragraph entitled "Hazardous Debris"), including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations. Material not regulated as solid waste are: nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.2 Debris

Non-hazardous solid material generated during the construction, demolition, or renovation of a structure which exceeds 2.5 inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.3 Hazardous Debris

As defined in paragraph entitled "Debris" of this section, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) per 40 CFR 261; or debris that exhibits a characteristic of hazardous waste per 40 CFR 261.

1.2.4 Chemical Wastes

This includes salts, acids, alkalies, herbicides, pesticides, and organic chemicals.

1.2.5 Garbage

Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.6 Hazardous Waste

Hazardous waste as defined in 40 CFR 261 or as defined by applicable State and local regulations.

1.2.7 Oily Waste

Petroleum products and bituminous materials.

1.2.8 Regulated Waste

Those solid waste that have specific additional Federal, state, or local controls for handling, storage, or disposal.

1.2.9 Hazardous Materials

Any material that is regulated as a hazardous material in accordance with 49 CFR 173, requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have the potential to meet the definition of a Hazardous Waste in accordance with 40 CFR 261. Throughout this specification, hazardous material includes hazardous chemicals.

1.3 SUBMITTALS

Submit the following in accordance with Section 01330, "Submittal Procedures."

SD-01 Preconstruction Submittals

Environmental protection plan

1.4 REPORTS

1.4.1 Regulatory Notification

The Contractor is responsible for all regulatory notification requirements in accordance with Federal, State and local regulations.

1.4.2 Solid Waste Disposal Report

Monthly the Contractor shall submit a solid waste disposal report to the Contracting Officer.

1.5 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during normal construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Comply with Federal, State, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

1.5.1 Contractor Liabilities for Environmental Protection

The Contractor is advised that this project and the station are subject to Federal, State, and local regulatory agency inspections to review compliance with environmental laws and regulations. The Contractor shall fully cooperate with any representative from any Federal, State or local regulatory agency who may visit the job site and shall provide immediate notification to the Contracting Officer, who shall accompany them on any subsequent site inspections. The Contractor shall complete, maintain, and make available to the Contracting Officer, station, or regulatory agency personnel all documentation relating to environmental compliance under applicable Federal, State and local laws and regulations. The Contractor shall immediately notify the Contracting Officer if a Notice of Violation (NOV) is issued to the Contractor.

1.6 UNFORESEEN HAZARDOUS OR REGULATED MATERIAL

If material that is not indicated in the contract documents is encountered that may be dangerous to human health upon disturbance during construction operations, stop that portion of work and notify the Contracting Officer immediately. Intent is to identify materials such as PCB, lead paint, mercury, petroleum products, and friable and nonfriable asbestos. Within 14 calendar days the Government will determine if the material is hazardous. If the material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If the material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to "FAR 52.243-4, Changes" and "FAR 52.236-2, Differing Site Conditions."

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition

upon completion of work. Confine construction activities to within the limits of the work indicated or specified.

3.1.1 Water Resources

3.1.1.1 Oily and Hazardous Substances

Prevent oil or hazardous substances from entering the ground, drainage areas, or navigable waters.

3.1.2 Fish and Wildlife Resources

Do not disturb fish and wildlife. Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as indicated or specified.

3.2 CONTROL AND DISPOSAL OF SOLID WASTES

Pick up solid wastes, and place in covered containers which are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Recycling is encouraged and can be coordinated with the Contracting Officer and the activity recycling coordinator. Remove all solid waste (including non-hazardous debris) from Government property and dispose off-site at an approved landfill. Solid waste disposal off-site must comply with most stringent local, State, and Federal requirements including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

3.2.1 Dumpsters

Contractor is responsible for providing trash dumpster. These dumpsters shall be Equip dumpsters with a secure cover. Keep cover closed at all times, except when being loaded with trash and debris. Locate dumpsters behind the construction fence or out of the public view. Empty site dumpsters at least once a week or as needed to keep the site free of debris and trash. If necessary, provide 55 gallon trash containers to collect debris in the construction site area. Locate the trash containers behind the construction fence or out of the public view. Empty trash containers at least once a day. For large demolitions, large dumpsters without lids are acceptable but should not have debris higher than the sides before emptying.

3.3 DUST CONTROL

Keep dust down at all times, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete.

-- End of Section --

SECTION 01 77 00

CLOSEOUT PROCEDURES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-1-2909	(2012) Engineering and Design -- Geospatial Data and Systems
ERDC/ITL TR-12-1	(2015) A/E/C Graphics Standard, Release 2.0
ERDC/ITL TR-12-6	(2015) A/E/C CAD Standard - Release 6.0

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up (i.e., red-line) drawings, maintained by the KTR on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the KO; design that is the responsibility of the KTR, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site and provide a copy to the COR monthly. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as convenience in the preparation of construction drawings and data for the referenced project. Any other use or reuse shall be at the sole risk of the KTR and without liability or legal exposure to the Government. The KTR must make no claim and waives to the fullest extent

permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The KTR must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the KTR hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The KTR is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the KTR of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the KTR uses, duplicates or modifies these electronic source drawing files for use in producing construction drawings and data related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.4 SUBMITTALS

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

1.4.1 SD-10 Operation and Maintenance Data

Equipment/Product Warranty List

Submit the above submittals within 14 calendar days following completion of work, unless noted otherwise.

1.4.2 SD-11 Closeout Submittals

As-Built Drawings (Provide monthly with pay application)

Record Drawings

Final Approved Shop Drawings

Submit the above submittals within 14 calendar days following completion of work, unless noted otherwise.

PART 2 PRODUCTS

2.1 AS-BUILT DRAWINGS

Provide electronic PDF "plots" of all contract drawings sheets associated with the as-built drawing submittal. Compile and organize the PDF set to match the contract drawings. Provide PDFs of the current working redlines

and/or markups complying with the as-builts drawing and markup requirements contained in this specification.

2.2 RECORD DRAWINGS

Provide record drawings in PDF and AutoCAD Release 2020 format.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site or red-lined PDF files. Submit As-Built Drawings monthly with pay applications.

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as follows:
 - (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.
- c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
- d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
- e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
- f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
- g. For deletions, cross out all features, data and captions that relate to that revision.

- h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
- i. Indicate one of the following when attaching a print or sketch to a markup print:
 - 1) Add an entire drawing to contract drawings
 - 2) Change the contract drawing to show
 - 3) Provided for reference only to further detail the initial design.
- j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content

Revise As-Built Drawings in accordance with ERDC/ITL TR-12-1 and ERDC/ITL TR-12-6. Keep these working as-built markup drawings current on a weekly basis and at least one set available on the jobsite at all times. Changes from the contract drawings which are made during construction or additional information which might be uncovered in the course of construction (including mislocated utilities encountered) must be accurately and neatly recorded as they occur by means of details and notes. Submit the working as-built markup drawings for approval prior to submission of each monthly pay estimate. For failure to maintain the working and final record drawings as specified herein, the KO may withhold 10 percent of the monthly progress payment until approval of updated drawings. Show on the as-built drawings, but not limited to, the following information:

- a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.
- b. The location and dimensions of any changes within the building structure.
- c. Layout and schematic drawings of electrical circuits and piping.
- d. Correct grade, elevations, cross sections, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.
- e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.
- f. The topography, inverted elevations and grades of drainage installed or affected as part of the project construction.
- g. Changes or Revisions which result from the final inspection.

- h. Where contract drawings or specifications present options show only the option selected for construction on the working as-built markup drawings.
- i. If borrowing material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Changes in location of equipment and architectural features.
- l. Actual location of anchors, construction and control joints, etc., in concrete.
- m. Unusual or uncharted obstructions that are encountered in the contract work area during construction.
- n. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.2 RECORD DRAWING FILES

If additional drawings are required, prepare them using the specified electronic file format applying [the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings. Accomplish additions and corrections to the contract drawings using CAD files. Provide all program files and hardware necessary to prepare final PDF record drawings. The COR will review final PDF record drawings for accuracy and return them to the KTR for required corrections, changes, additions, and deletions.

3.3 FINAL RECORD DRAWING PACKAGE

Submit the final record PDF and CAD drawings package for the entire project within 20 days of substantial completion of all phases of work. Submit one set of ANSI D size PDF and CAD files, two sets of ANSI D size prints and one set of the approved working record drawings. The package must be complete in all details and identical in form and function to the contract drawing files supplied by the Government.

3.5 CLEANUP

Leave premises "broom clean." Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site.

REPLACE MEDIUM VOLTAGE CUTOFF SWITCHES
CG BASE CAPE COD

PSN 16677450

-- End of Section --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

1.1 SUBMISSION OF OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data/Manuals which are specifically applicable to this contract and a complete and concise depiction of the provided equipment or product. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this section and Section 01330, "Submittal Procedures."

1.1.1 Quantity

Submit three sets of the supplier/manufacturers' O&M information specified herein for the components, assemblies, subassemblies, attachments, and accessories. The items for which O&M Data/Manuals are required are listed in the technical sections which specifies those particular items.

1.1.2 Package Quality

Documents must be fully legible. Poor quality copies and material with hole punches obliterating the text or drawings will not be accepted.

1.1.3 Package Content

Data package content shall be as shown in the paragraph titled "Schedule of Operation and Maintenance Data Packages." For each product, system, or component piece of equipment requiring submission of O&M Data, submit the Data Package specified in the individual technical section.

1.1.4 Delivery

Submit O&M Data Manuals to the Contracting Officer for review and acceptance; submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.

- a. In the event the Contractor fails to deliver O&M Data/Manuals within the time limits set forth above, the Contracting Officer may withhold from progress payments 50 percent of the price of the item with which such O&M Data/Manuals are associated.

1.1.5 Changes to Submittals

Manufacturer-originated changes or revisions to submitted data shall be furnished by the Contractor if a component of an item is so affected subsequent to acceptance of the O&M Data. Changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data, shall be submitted by the Contractor within 30 calendar days of the notification of this change requirement.

1.2 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

1.2.1 Operating Instructions

Include specific instructions, procedures, safety precautions, and illustrations for the following phases of operation:

1.2.1.1 Normal Operations

Provide narrative description of normal operating procedures. Include control diagrams with data to explain operation and control of systems and specific equipment.

1.2.1.2 Emergency Operations

Include emergency procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Include emergency shutdown instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance on emergency operations of all utility systems including valve locations and portions of systems controlled.

1.2.1.3 Operator Service Requirements

Include instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and gage reading recording.

1.2.1.4 Environmental Conditions

Include a list of environmental conditions (temperature, humidity, and other relevant data) which are best suited for each product or piece of equipment and describe conditions under which equipment should not be allowed to run.

1.2.2 Preventive Maintenance

Include the following information for preventive and scheduled maintenance to minimize corrective maintenance and repair.

1.2.2.1 Preventive Maintenance Plan and Schedule

Include manufacturer's schedule for routine preventive maintenance, inspections, tests and adjustments required to ensure proper and economical operation and to minimize corrective maintenance and repair. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

1.2.3 Troubleshooting Guides and Diagnostic Techniques

Include step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.2.4 Wiring Diagrams and Control Diagrams

Wiring diagrams and control diagrams shall be point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation numbering.

1.2.5 Maintenance Procedures

Include instructions and list tools required to restore product or equipment to proper condition or operating standards.

1.2.6 Removal and Replacement Instructions

Include step-by-step procedures and list required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments.

1.2.7 Spare Parts and Supply Lists

Include lists of spare parts and supplies required for maintenance and repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead time to obtain.

1.2.8 Parts Identification

Provide identification and coverage for all parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing shall show the index, reference, or key number which will cross-reference the illustrated part to the listed part. Parts shown in the listings shall be grouped by components, assemblies, and subassemblies. Parts data may cover more than one model or series of equipment. components, assemblies, subassemblies, attachments, or accessories, such as a master parts catalog, in accordance with the manufacturer's standard commercial practice.

1.2.8.1 Warranty Information

List and explain the various warranties and include the servicing and technical precautions prescribed by the manufacturers or contract documents to keep warranties in force. Include warranty information for primary components.

1.2.8.2 Personnel Training Requirements

Provide information available from the manufacturers to use in training designated personnel to operate and maintain the equipment and systems properly.

1.2.8.3 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components.

1.2.8.4 Contractor Information

Provide a list that includes the name, address, and telephone number of the General Contractor and each subcontractor installing the product or equipment. Include local representatives and service organizations most convenient to the project site. Provide the name, address, and telephone number of the product or equipment manufacturers.

1.3 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Furnish the O&M Data Packages specified in individual technical sections. The required information for each O&M Data Package is as follows:

1.3.1 Data Package 1

- a. Safety precautions
- b. Maintenance procedures
- c. Warranty information
- d. Contractor information

1.3.2 Data Package 2

- a. Safety precautions
- b. Normal operations
- c. Environmental conditions
- d. Preventive maintenance plan and schedule
- e. Maintenance procedures
- f. Removal and replacement instructions

- g. Spare parts and supply list
- h. Parts identification
- i. Warranty information
- j. Contractor information

1.3.3 Data Package 3

- a. Safety precautions
- b. Normal operations
- c. Emergency operations
- d. Environmental conditions
- e. Lubrication data
- f. Preventive maintenance plan and schedule
- g. Troubleshooting guides and diagnostic techniques
- h. Wiring diagrams and control diagrams
- i. Maintenance and repair procedures
- j. Removal and replacement instructions
- k. Spare parts and supply list
- l. Parts identification
- m. Warranty information
- n. Testing equipment and special tool information
- o. Contractor information

1.3.4 Data Package 4

- a. Safety precautions
- b. Operator prestart
- c. Startup, shutdown, and post shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Operator service requirements
- g. Environmental conditions

- h. Lubrication data
- i. Preventive maintenance plan and schedule
- j. Troubleshooting guides and diagnostic techniques
- k. Wiring diagrams and control diagrams
- l. Maintenance and repair procedures
- m. Removal and replacement instructions
- n. Spare parts and supply list
- o. Corrective maintenance man-hours
- p. Parts identification
- q. Warranty information
- r. Personnel training requirements
- s. Testing equipment and special tool information
- t. Contractor information

1.3.5 Data Package 5

- a. Safety precautions
- b. Operator prestart
- c. Start-up, shutdown, and post shutdown procedures
- d. Normal operations
- e. Environmental conditions
- f. Preventive maintenance plan and schedule
- g. Troubleshooting guides and diagnostic techniques
- h. Wiring and control diagrams
- i. Maintenance and repair procedures
- j. Spare parts and supply list
- k. Testing equipments and special tools
- l. Warranty information
- m. Contractor information

PART 2 PRODUCTS

Not used.

REPLACE MEDIUM VOLTAGE CUTOFF SWITCHES
CG BASE CAPE COD

PSN 16677450

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 80 00

FACILITIES PREVENTATIVE MAINTENANCE PROGRAM

PART 1 GENERAL

1.1 GENERAL

The intent of this section is for the Contractor to complete the Equipment Enrollment Form spreadsheet (Attachment #1) for systems that have been installed or demolished under this project and are listed in the USCG Approved Equipment Enrollment Catalog to assist the Coast Guard's Facilities Preventative Maintenance Program (FPMP).

At a minimum, the following items require UNIFORMAT II Level 4 designations:

- a. Electrical Systems

Spec Attachments:

- a. USCG Approved Equipment Enrollment Catalog (Attachment #3, Provided electronically by USCG COR within 14-days of requesting)
- b. Sample Job Plan (Attachment #2, Provided Electronically by USCG COR within 14-days of requesting)
- c. Coast Guard Maintained Equipment Enrollment Form (Attachment #1, Provided Electronically by USCG COR within 14-days of requesting)

1.2 OVERVIEW

This section describes the requirements for:

- a. Facilities Preventative Maintenance Program.

Refer to Section 01 78 23 OPERATION AND MAINTENANCE DATA for operation and maintenance data requirements.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1557 (2015) Standard Classification for Building Elements and Related Sitework - UNIFORMAT II

1.4 SUBMITTALS

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

1.3.1 SD-12 Submittals

Facilities Preventative Maintenance Program (FPMP)

- (a) Draft Equipment Enrollment Form (EEF) of equipment inventory and PM procedures.
- (b) Corrected Equipment Enrollment Form of equipment inventory and PM procedures.
- (c) Final Equipment Enrollment Form of equipment inventory and PM procedures.
- (d) Draft O&M items requiring new SAM's Job Plan data input.
- (e) Corrected O&M items requiring new SAM's Job Plan data input.
- (f) Final O&M items requiring new SAM's Job Plan data input.

1.3.1.1 Submissions

- (a) Submit two (2) Draft copies of EEF, PM procedures, and SAM's Job Plan for review by the COR and USCG SAM POC.
- (b) Submit two (2) Corrected copies of EEF, PM procedures, and SAM's Job Plan for review by the COR and USCG SAM POC.
- (c) Submit two (2) Final copies of EEF, PM procedures, and SAM's Job Plan for review by the COR and USCG SAM POC.

PART 2 PRODUCTS

2.1 FACILITIES PREVENTATIVE MAINTENANCE PROGRAM (FPMP)

The Contractor shall develop and submit an FPMP for all newly constructed or installed infrastructure that accurately identifies the preventive maintenance required to professionally maintain each system or piece of equipment provided in this contract. Work includes, but is not limited to: identifying and documenting all facilities items, systems, equipment, and components requiring maintenance and/or special operation; identifying and documenting the preventative maintenance requirement; and supplying this information on a government provided Excel Spreadsheet. Contractor shall also list equipment that has been demolished or removed due to the project. The Government will take the Contractor supplied information and input it into the Coast Guard Shore Asset Management (SAM) system.

2.1.1 USCG Approved Equipment Enrollment Catalog

The Catalog provided is the most current list of equipment that is maintained by the Coast Guard in SAM. The items that shall be listed on the Equipment Enrollment Form are identified in attachment 3 and highlighted in yellow within the catalog electronic spreadsheet. For each piece of equipment provided or installed as part of this contract, complete the "Equipment Enrollment Form". The purpose of the USCG Approved Equipment Enrollment Catalog is as follows:

- a. Lists the equipment that the Coast Guard performs preventive maintenance per standard scheduling and job plans.
- b. Catalogs the equipment according ASTM E1557 "Standard Classification for Building Elements and Related Site work - UNIFORMAT II, level 4.
- c. Assigns Preventive Maintenance Procedures to the equipment per the "Maintenance & Repair" tab and "Preventive Maintenance Tab" of the RS MEANS Company, "FACILITIES MAINTENANCE AND REPAIR COST DATA" publication (not provided); and
- d. Assigns the SAM "Job Plan" to the respective RS Means Preventative Maintenance Procedure in a one-to-one database context.

2.1.1.2 "Equipment Enrollment Form" Requirements

2.1.1.2.1 Form Fields

The following fields are listed on the form and shall be completely filled out except where otherwise noted on the Equipment Enrollment Form. The actual equipment attribute list below may change slightly prior to the actual start of this enrollment task. Column references identified refer back to the Equipment Enrollment Catalog spreadsheet.

- a. UNIFORMAT II Level IV Classification (column K)
- b. Equipment Description (column AC).
- c. Physical Location, broken down by Bldg Number, Floor, and Room #.
- d. Manufacturer Name.
- e. Model Number.
- f. Serial #.
- g. Installation Date.
- h. Purchase Price (Cost of equipment, labor, shipping).
- i. Replacement Costs (Cost of equipment only).
- j. Warranty Expiration Date.
- k. Equipment Attributes (Name Plate information typically indicating Size, Flow, Volume, Pressure, etc.). Reference columns AJ through AV.
- l. Standard Coast Guard SAM "Job Plan" Number (column U).
- m. PM Job Plan Frequency (column V).

2.1.1.2.2 SAM Building Numbers

Only equipment from a single SAM Location/Building # is allowed per "Equipment Enrollment Form". Allow the Coast Guard 14 days to provide a listing of all applicable SAM Location/Building #'s.

2.1.1.2.3 Equipment Identified for Maintenance

Equipment identified for maintenance by O&M manuals but not listed on the USCG Equipment Enrollment Catalog shall be cataloged per ASTM E1557 and listed on the equipment enrollment form. If the equipment is not listed in ASTM E1557, consult the Coast Guard for the proper naming convention.

2.1.1.2.3.1 RS Means Maintenance Procedures

Contractor shall review RS Means Maintenance procedures, contained within the RS MEANS FACILITIES MAINTENANCE AND REPAIR COST DATA publication, for each piece of equipment that it lists on the "Equipment Enrollment Form" but not found in the USCG Equipment Enrollment Catalog and provide new "Job Plan" in a separate document for Coast Guard modification of the SAM database. Attachment #2 provides a sample/template job plan.

As part of providing a new "job plan", the Contractor shall furnish the labor-hours, equipment cost, material cost, total in-house field, and total with overhead and profit (O&P) fields for each PM activity similar to what is provided by the RS MEANS Company's, "FACILITIES MAINTENANCE AND REPAIR COST DATA" publication.

2.1.1.2.3.2 Job Plan

If RS MEANS does not have a preventative maintenance plan for a specific piece of equipment, provide a detailed Job Plan using manufacturer's recommendations and sound engineering practice using template job plan. Show associated frequencies when job plan is to be performed and include detailed preventative maintenance procedures such as inspections, tests, and adjustments, required to ensure proper and economical operation and minimize corrective maintenance. Detailed job plans include the following:

- a. Safety instruction and precautions.
- b. Including lock out/tag out precautions.
- c. Required skill level.
- d. Number of personnel needed.
- e. Frequency of performing the job plan.
- f. Special tools needed.
- g. Parts needed.
- h. Estimated time required to complete the task.
- i. Lubrication schedules indicating types, grades, and capacities.

2.1.2.4 Deliverables include:

- a. Electronic Excel files and formatted hard copy print-out of completed Equipment Enrollment Form (Coast Guard will complete data entry into SAM).
- b. Electronic files and formatted hard copy print out of any newly created "Job Plan" for equipment applicable under paragraph entitled "Equipment Identified for Maintenance."

2.1.2.5 Number of Copies

Contractor shall submit three (3) deliverables to include a DRAFT, Corrected, and FINAL of the FPMP to the Coast Guard for review, comment, and acceptance.

2.1.2.6 Submissions

Submit the DRAFT FPMP four (4) months prior to construction completion date (CCD). Government review of the DRAFT will be 2 weeks. Submit the Corrected FPMP no later than 1 month after receiving the Government's comments on the Draft submission. Submit the FINAL FPMP no later than 2 weeks after receiving the Government's comments on the Corrected submission.

2.1.3 Qualifications

The FPMP Preparer shall have the following qualifications:

- a. Possess multi-discipline technical knowledge of the operations and maintenance of building systems.
- b. Experience with the type of systems that are identified in the project specification and capable of augmenting manufacturer's information to clarify operations instructions.
- c. Experience preparing detailed operations and maintenance manuals for facilities of equal size and complexity as required by this contract.
- d. Ability to prepare spreadsheets to be manually loaded into the Coast Guard's SAM database.
- e. Experience in presenting training classes and coordinating a team of manufacturer's representatives to provide training for facility

users and maintenance personnel.

PART 3 EXECUTION

Contractor shall provide all the requirements listed above.

-- End of Section --